

HDOC DAY-33

Title: Verification of Associative Properties of Matrix Addition and Multiplication

1. Problem Statement

Write a C program to perform the following menu-driven operations using separate functions:

1. Read three variable-length 2D integer arrays A , B , and C , each of dimensions $(m \times n)$, and verify the associative property of matrix addition: $(A + B) + C = A + (B + C)$.
2. Read three variable-length 2D integer arrays with the following dimensions: $A (m \times n)$, $B (n \times p)$, $C (p \times q)$. Verify the associative property of matrix multiplication: $(A \times B) \times C = A \times (B \times C)$.
3. Exit.

Note: For each operation, display the matrices obtained on both sides of the equation and state whether the associative property is verified.

2. Theoretical Background & Mathematical Proof

1. Associative Property of Matrix Addition:

For any three matrices $A, B, C \in \mathbb{R}^{m \times n}$, element-wise addition satisfies $(a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij})$ due to the associative law of real numbers. Thus, $(A + B) + C = A + (B + C)$.

2. Associative Property of Matrix Multiplication:

For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, and $C \in \mathbb{R}^{p \times q}$, the entry at position (i, l) in $[(AB)C]$ is given by:

$$\begin{aligned} [(AB)C]_{il} &= \sum_{k=1}^p \left(\sum_{j=1}^n a_{ij} b_{jk} \right) c_{kl} = \sum_{j=1}^n a_{ij} \left(\sum_{k=1}^p b_{jk} c_{kl} \right) \\ &= [A(BC)]_{il} \end{aligned}$$

Hence, matrix multiplication is strictly associative regardless of matrix entries.

3. Test Case Table

Test Case	Option	Input Dimensions & Matrices	LHS Computation	RHS Computation	Status
1	Addition Associativity	$m = 2, n = 2$			

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}, C = \begin{bmatrix} 9 & 1 \\ 2 & 3 \end{bmatrix} \mid (A + B) + C =$
 $\begin{bmatrix} 15 & 9 \\ 12 & 15 \end{bmatrix} \mid A + (B + C) = \begin{bmatrix} 15 & 9 \\ 12 & 15 \end{bmatrix} \mid \text{Verified} \mid \mid \mathbf{2} \mid$
 Multiplication Associativity $\mid m = 2, n = 2, p = 2, q = 2$

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mid (A \times B) \times C =$
 $\begin{bmatrix} 4 & 10 \\ 10 & 22 \end{bmatrix} \mid A \times (B \times C) = \begin{bmatrix} 4 & 10 \\ 10 & 22 \end{bmatrix} \mid \text{Verified} \mid \mid \mathbf{3} \mid \text{Exit} \mid$
 Choice 3 $\mid$ Exit system call $\mid$ Terminated cleanly $\mid$ Pass $\mid$

4. Source Code Listing

```

#include
#include

void read_matrix(int rows, int cols, int mat[rows][cols], const char
*name) {
    for (int i = 0; i < rows; i++)
        for (int j = 0; j < cols; j++)
            scanf("%d", &mat[i][j]);
}

void print_matrix(int rows, int cols, int mat[rows][cols]) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) printf("%d\t", mat[i][j]);
        printf("\n");
    }
}

```

```

bool are_matrices_equal(int rows, int cols, int mat1[rows][cols], int
mat2[rows][cols]) {
    for (int i = 0; i < rows; i++)
        for (int j = 0; j < cols; j++)
            if (mat1[i][j] != mat2[i][j]) return false;
    return true;
}

void add_matrices(int rows, int cols, int A[rows][cols], int B[rows][cols],
int res[rows][cols]) {
    for (int i = 0; i < rows; i++)
        for (int j = 0; j < cols; j++)
            res[i][j] = A[i][j] + B[i][j];
}

void multiply_matrices(int r1, int c1, int A[r1][c1], int r2, int c2, int
B[r2][c2], int res[r1][c2]) {
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c2; j++) {
            res[i][j] = 0;
            for (int k = 0; k < c1; k++) res[i][j] += A[i][k] * B[k][j];
        }
    }
}

void verify_addition_associative(void) {
    int m, n;
    printf("Enter dimensions m and n: "); scanf("%d %d", &m, &n);
    int A[m][n], B[m][n], C[m][n], A_plus_B[m][n], B_plus_C[m][n],
LHS[m][n], RHS[m][n];

    read_matrix(m, n, A, "A"); read_matrix(m, n, B, "B"); read_matrix(m, n, C,
"C");

    add_matrices(m, n, A, B, A_plus_B); add_matrices(m, n, A_plus_B, C,
LHS);
    add_matrices(m, n, B, C, B_plus_C); add_matrices(m, n, A, B_plus_C,
RHS);

    printf("\nLHS ((A+B)+C):\n"); print_matrix(m, n, LHS);

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printf("\nRHS (A+(B+C)):\n"); print_matrix(m, n, RHS);

if (are_matrices_equal(m, n, LHS, RHS))
    printf("\nVERIFIED: Associative Property of Matrix Addition Holds.\n");
}

void verify_multiplication_associative(void) {
    int m, n, p, q, temp;
    printf("Enter m n for A: "); scanf("%d %d", &m, &n);
    printf("Enter n p for B: "); scanf("%d %d", &temp, &p);
    printf("Enter p q for C: "); scanf("%d %d", &temp, &q);

    int A[m][n], B[n][p], C[p][q], AB[m][p], BC[n][q], LHS[m][q], RHS[m][q];
    read_matrix(m, n, A, "A"); read_matrix(n, p, B, "B"); read_matrix(p, q, C,
"C");

    multiply_matrices(m, n, A, n, p, B, AB); multiply_matrices(m, p, AB, p, q,
C, LHS);
    multiply_matrices(n, p, B, p, q, C, BC); multiply_matrices(m, n, A, n, q,
BC, RHS);

    printf("\nLHS ((A*B)*C):\n"); print_matrix(m, q, LHS);
    printf("\nRHS (A*(B*C)):\n"); print_matrix(m, q, RHS);

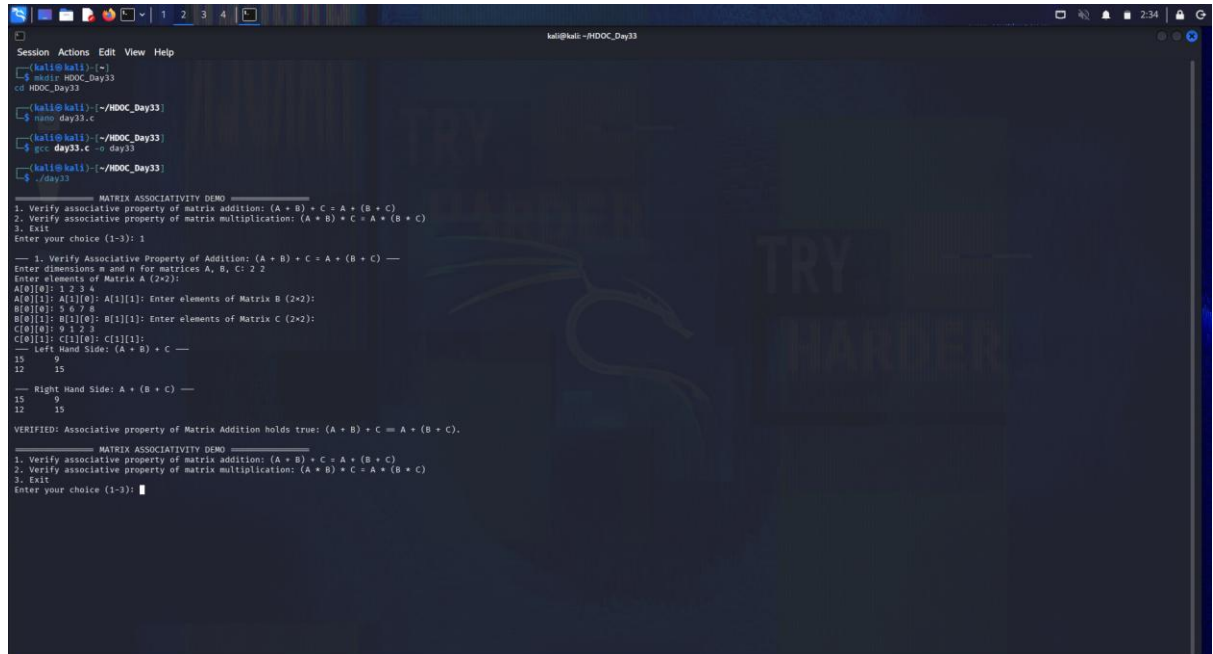
    if (are_matrices_equal(m, q, LHS, RHS))
        printf("\nVERIFIED: Associative Property of Matrix Multiplication
Holds.\n");
}

int main(void) {
    int choice;
    do {
        printf("\n1. Verify Addition Associativity\n2. Verify Multiplication
Associativity\n3. Exit\nChoice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: verify_addition_associative(); break;
            case 2: verify_multiplication_associative(); break;
            case 3: printf("Exiting...\n"); break;
        }
    }
}

```

```
} while (choice != 3);  
return 0;  
}
```

4. Execution Screenshots



```
kali@kali:~/HDOOC_Day33  
$ gcc day33.c  
$ ./day33  
===== MATRIX ASSOCIATIVITY DEMO =====  
1. Verify associative property of matrix addition: (A + B) + C = A + (B + C)  
2. Verify associative property of matrix multiplication: (A * B) * C = A * (B * C)  
3. Exit  
Enter your choice (1-3): 1  
----- 1. Verify Associative Property of Addition: (A + B) + C = A + (B + C) -----  
Enter dimensions n and m for matrices A, B, C: 2 2  
Enter elements of Matrix A (2x2):  
A[0][0]: 1 2 3 4  
A[0][1]: A[1][0]: A[1][1]: Enter elements of Matrix B (2x2):  
B[0][0]: 5 6 7 8  
B[0][1]: B[1][0]: B[1][1]: Enter elements of Matrix C (2x2):  
C[0][0]: 9 1 2 3  
C[0][1]: C[1][0]: C[1][1]:  
----- Left Hand Side: (A + B) + C -----  
15 9  
12 15  
----- Right Hand Side: A + (B + C) -----  
15 9  
12 15  
VERIFIED: Associative property of Matrix Addition holds true: (A + B) + C = A + (B + C).  
===== MATRIX ASSOCIATIVITY DEMO =====  
1. Verify associative property of matrix addition: (A + B) + C = A + (B + C)  
2. Verify associative property of matrix multiplication: (A * B) * C = A * (B * C)  
3. Exit  
Enter your choice (1-3):
```